

Weekly Topical Worksheet Pack**CANDIDATE****Urwah****SUBJECT****Chemistry -- Cambridge International A Level (9701)****TOTAL MARKS (WEEK)****520 marks****HOW TO USE THIS PACK**

One short set of questions per session, matching that day's topic from the schedule.

Answer in the spaces / on the lines provided. Marks for each part are shown in [brackets].

Mark schemes for the whole week are collected at the end of the pack.

These are original questions written to match Cambridge's exact style and command words — not reproductions of real past paper text.

THIS WEEK AT A GLANCE

TOPIC 1 — TOPIC 1: ATOMIC STRUCTURE (50 marks)

TOPIC 2 — TOPIC 2: ATOMS, MOLECULES AND STOICHIOMETRY (59 marks)

TOPIC 3 — TOPIC 3: CHEMICAL BONDING (58 marks)

TOPIC 4 — TOPIC 4: STATES OF MATTER (50 marks)

TOPIC 5 — TOPIC 5: CHEMICAL ENERGETICS (75 marks)

TOPIC 6 — TOPIC 6: ELECTROCHEMISTRY (44 marks)

TOPIC 7 — TOPIC 7: EQUILIBRIA (70 marks)

TOPIC 8 — TOPIC 8: REACTION KINETICS (44 marks)

TOPIC 9 — TOPIC 9: THE PERIODIC TABLE: CHEMICAL PERIODICITY (70 marks)

MENTORA ACADEMY · EXCELLENCE IN EDUCATION.**TOPICAL PRACTICE — WEEK 1****Physical Chemistry -- Topics 1-9: Atomic Structure to Reaction Kinetics**

AS Level Chemistry 9701 -- 2025-2027 Syllabus

Question Pack

TOPIC 1 – TOPIC 1: ATOMIC STRUCTURE

1 Atomic Structure – 9701/11/M/J/20/Q1 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

2 Atomic Structure – 9701/11/M/J/20/Q2 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

3 Atomic Structure – 9701/11/M/J/20/Q3 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

4 Atomic Structure – 9701/11/M/J/20/Q4 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

5 Atomic Structure – 9701/11/M/J/20/Q5 Syllabus 1.1

Which statement about the particles within an atom is correct?

(a)

What is the relative mass and charge of a proton?

A Mass 1, Charge 0

B Mass 1, Charge +1

C Mass 1/1836, Charge -1

D Mass 0, Charge +1

[1]

6 Isotopes – 9701/22/O/N/21/Q1(a) Syllabus 1.2

(a)

Define the term isotope.

[2]

Question Pack

- 7 Isotopes — 9701/22/O/N/21/Q2(a)** Syllabus 1.2
(a) Define the term isotope. [2]
- 8 Isotopes — 9701/22/O/N/21/Q3(a)** Syllabus 1.2
(a) Define the term isotope. [2]
- 9 Isotopes — 9701/22/O/N/21/Q4(a)** Syllabus 1.2
(a) Define the term isotope. [2]
- 10 Isotopes — 9701/22/O/N/21/Q5(a)** Syllabus 1.2
(a) Define the term isotope. [2]
- 11 Ionisation Energy — 9701/21/M/J/19/Q1** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]
- 12 Ionisation Energy — 9701/21/M/J/19/Q2** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]
- 13 Ionisation Energy — 9701/21/M/J/19/Q3** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]
- 14 Ionisation Energy — 9701/21/M/J/19/Q4** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]
- 15 Ionisation Energy — 9701/21/M/J/19/Q5** Syllabus 1.4
[DIAGRAM: Graph showing successive ionisation energies of element X]
(a) Explain the general increase in successive ionisation energies of element X. [2]
(b) Deduce the group number of element X. [1]

Question Pack

- 16 Orbitals — 9701/13/O/N/22/Q1** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 17 Orbitals — 9701/13/O/N/22/Q2** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 18 Orbitals — 9701/13/O/N/22/Q3** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 19 Orbitals — 9701/13/O/N/22/Q4** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 20 Orbitals — 9701/13/O/N/22/Q5** Syllabus 1.3 [1]
(a) Which electron configuration represents an element in Group 15?
A $1s^2 2s^2 2p^6 3s^2 3p^3$
B $1s^2 2s^2 2p^6 3s^2$
C $1s^2 2s^2 2p^6$
D $1s^2 2s^2 2p^5$
- 21 Atomic Radius — 9701/22/F/M/23/Q1(a)** Syllabus 1.1 [3]
(a) Explain why the atomic radius decreases across Period 3 from Na to Cl.
- 22 Atomic Radius — 9701/22/F/M/23/Q2(a)** Syllabus 1.1 [3]
(a) Explain why the atomic radius decreases across Period 3 from Na to Cl.
- 23 Atomic Radius — 9701/22/F/M/23/Q3(a)** Syllabus 1.1 [3]
(a) Explain why the atomic radius decreases across Period 3 from Na to Cl.

Question Pack**24 Atomic Radius — 9701/22/F/M/23/Q4(a)** Syllabus 1.1

- (a) Explain why the atomic radius decreases across Period 3 from Na to Cl. **[3]**

25 Atomic Radius — 9701/22/F/M/23/Q5(a) Syllabus 1.1

- (a) Explain why the atomic radius decreases across Period 3 from Na to Cl. **[3]**

Question Pack

TOPIC 2 – TOPIC 2: ATOMS, MOLECULES AND STOICHIOMETRY

- 26 Moles – 9701/21/M/J/21/Q1(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 27 Moles – 9701/21/M/J/21/Q2(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 28 Moles – 9701/21/M/J/21/Q3(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 29 Moles – 9701/21/M/J/21/Q4(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 30 Moles – 9701/21/M/J/21/Q5(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 31 Moles – 9701/21/M/J/21/Q6(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 32 Moles – 9701/21/M/J/21/Q7(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 33 Moles – 9701/21/M/J/21/Q8(b)** Syllabus 2.2
(a) Calculate the number of moles of CO₂ in 1.20 dm³ of gas at room temperature and pressure. [2]
- 34 Titration – 9701/23/O/N/20/Q1(c)** Syllabus 2.4
25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.
(a) Calculate the concentration of the H₂SO₄. [3]
- 35 Titration – 9701/23/O/N/20/Q2(c)** Syllabus 2.4
25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.
(a) Calculate the concentration of the H₂SO₄. [3]
- 36 Titration – 9701/23/O/N/20/Q3(c)** Syllabus 2.4
25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.
(a) Calculate the concentration of the H₂SO₄. [3]
- 37 Titration – 9701/23/O/N/20/Q4(c)** Syllabus 2.4
25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.
(a) Calculate the concentration of the H₂SO₄. [3]

Question Pack

38 Titration – 9701/23/O/N/20/Q5(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

39 Titration – 9701/23/O/N/20/Q6(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

40 Titration – 9701/23/O/N/20/Q7(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

41 Titration – 9701/23/O/N/20/Q8(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

42 Titration – 9701/23/O/N/20/Q9(c) Syllabus 2.4

25.0 cm³ of 0.100 mol dm⁻³ NaOH was neutralised by 20.0 cm³ of H₂SO₄.

(a)

Calculate the concentration of the H₂SO₄.

[3]

43 Empirical Formula – 9701/22/F/M/22/Q1(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

44 Empirical Formula – 9701/22/F/M/22/Q2(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

45 Empirical Formula – 9701/22/F/M/22/Q3(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

46 Empirical Formula – 9701/22/F/M/22/Q4(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

47 Empirical Formula – 9701/22/F/M/22/Q5(d) Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

Question Pack

48 Empirical Formula – 9701/22/F/M/22/Q6(d)

Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]**49 Empirical Formula – 9701/22/F/M/22/Q7(d)**

Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]**50 Empirical Formula – 9701/22/F/M/22/Q8(d)**

Syllabus 2.3

A hydrocarbon contains 85.7% carbon by mass.

(a)

Calculate its empirical formula.

[2]

Question Pack

TOPIC 3 — TOPIC 3: CHEMICAL BONDING

- 51 VSEPR — 9701/21/M/J/19/Q1(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 52 VSEPR — 9701/21/M/J/19/Q2(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 53 VSEPR — 9701/21/M/J/19/Q3(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 54 VSEPR — 9701/21/M/J/19/Q4(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 55 VSEPR — 9701/21/M/J/19/Q5(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 56 VSEPR — 9701/21/M/J/19/Q6(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 57 VSEPR — 9701/21/M/J/19/Q7(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 58 VSEPR — 9701/21/M/J/19/Q8(a)** Syllabus 3.5
(a) State the shape of the ammonia molecule, NH_3 , and its bond angle. [2]
- 59 Dot and Cross — 9701/22/O/N/23/Q1(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]
- 60 Dot and Cross — 9701/22/O/N/23/Q2(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]
- 61 Dot and Cross — 9701/22/O/N/23/Q3(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]
- 62 Dot and Cross — 9701/22/O/N/23/Q4(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]
- 63 Dot and Cross — 9701/22/O/N/23/Q5(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO_2] Describe the bonding in CO_2 . [2]

Question Pack

- 64 Dot and Cross — 9701/22/O/N/23/Q6(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO₂] Describe the bonding in CO₂. [2]
- 65 Dot and Cross — 9701/22/O/N/23/Q7(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO₂] Describe the bonding in CO₂. [2]
- 66 Dot and Cross — 9701/22/O/N/23/Q8(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO₂] Describe the bonding in CO₂. [2]
- 67 Dot and Cross — 9701/22/O/N/23/Q9(b)** Syllabus 3.7
(a) [DIAGRAM: Draw a dot-and-cross diagram of CO₂] Describe the bonding in CO₂. [2]
- 68 Intermolecular Forces — 9701/23/F/M/21/Q1(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 69 Intermolecular Forces — 9701/23/F/M/21/Q2(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 70 Intermolecular Forces — 9701/23/F/M/21/Q3(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 71 Intermolecular Forces — 9701/23/F/M/21/Q4(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 72 Intermolecular Forces — 9701/23/F/M/21/Q5(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 73 Intermolecular Forces — 9701/23/F/M/21/Q6(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 74 Intermolecular Forces — 9701/23/F/M/21/Q7(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]
- 75 Intermolecular Forces — 9701/23/F/M/21/Q8(c)** Syllabus 3.6
(a) Explain why water has a higher boiling point than hydrogen sulfide. [3]

Question Pack

TOPIC 4 — TOPIC 4: STATES OF MATTER

76 Ideal Gas — 9701/21/M/J/20/Q1(a) Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**77 Ideal Gas — 9701/21/M/J/20/Q2(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**78 Ideal Gas — 9701/21/M/J/20/Q3(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**79 Ideal Gas — 9701/21/M/J/20/Q4(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**80 Ideal Gas — 9701/21/M/J/20/Q5(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**81 Ideal Gas — 9701/21/M/J/20/Q6(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**82 Ideal Gas — 9701/21/M/J/20/Q7(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**83 Ideal Gas — 9701/21/M/J/20/Q8(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**84 Ideal Gas — 9701/21/M/J/20/Q9(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]**85 Ideal Gas — 9701/21/M/J/20/Q10(a)** Syllabus 4.1

A gas has a volume of 0.050 m³ at a pressure of 1.01 x 10⁵ Pa and a temperature of 300 K.

- (a) Calculate the number of moles of gas present. ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)

[3]

Question Pack

- 86 Real Gases — 9701/22/O/N/22/Q1(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 87 Real Gases — 9701/22/O/N/22/Q2(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 88 Real Gases — 9701/22/O/N/22/Q3(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 89 Real Gases — 9701/22/O/N/22/Q4(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 90 Real Gases — 9701/22/O/N/22/Q5(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 91 Real Gases — 9701/22/O/N/22/Q6(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 92 Real Gases — 9701/22/O/N/22/Q7(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 93 Real Gases — 9701/22/O/N/22/Q8(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 94 Real Gases — 9701/22/O/N/22/Q9(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]
- 95 Real Gases — 9701/22/O/N/22/Q10(b)** Syllabus 4.1
(a) State two conditions under which a real gas behaves most like an ideal gas. [2]

Question Pack

TOPIC 5 — TOPIC 5: CHEMICAL ENERGETICS

- 96 Enthalpy — 9701/22/M/J/22/Q1(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 97 Enthalpy — 9701/22/M/J/22/Q2(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 98 Enthalpy — 9701/22/M/J/22/Q3(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 99 Enthalpy — 9701/22/M/J/22/Q4(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 100 Enthalpy — 9701/22/M/J/22/Q5(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 101 Enthalpy — 9701/22/M/J/22/Q6(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 102 Enthalpy — 9701/22/M/J/22/Q7(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 103 Enthalpy — 9701/22/M/J/22/Q8(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 104 Enthalpy — 9701/22/M/J/22/Q9(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 105 Enthalpy — 9701/22/M/J/22/Q10(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 106 Enthalpy — 9701/22/M/J/22/Q11(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 107 Enthalpy — 9701/22/M/J/22/Q12(a)** Syllabus 5.1
(a) Define standard enthalpy change of formation. [3]
- 108 Hess Law — 9701/21/O/N/19/Q1(b)** Syllabus 5.2
[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]
(a) Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. [3]

Question Pack

109 Hess Law — 9701/21/O/N/19/Q2(b) Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****110 Hess Law — 9701/21/O/N/19/Q3(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****111 Hess Law — 9701/21/O/N/19/Q4(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****112 Hess Law — 9701/21/O/N/19/Q5(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****113 Hess Law — 9701/21/O/N/19/Q6(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****114 Hess Law — 9701/21/O/N/19/Q7(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****115 Hess Law — 9701/21/O/N/19/Q8(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****116 Hess Law — 9701/21/O/N/19/Q9(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]****117 Hess Law — 9701/21/O/N/19/Q10(b)** Syllabus 5.2[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]**(a)**Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹. **[3]**

Question Pack

118 Hess Law — 9701/21/O/N/19/Q11(b) Syllabus 5.2

[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]

(a)

Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹.

[3]

119 Hess Law — 9701/21/O/N/19/Q12(b) Syllabus 5.2

[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]

(a)

Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹.

[3]

120 Hess Law — 9701/21/O/N/19/Q13(b) Syllabus 5.2

[DIAGRAM: Hess cycle showing formation of CH₄ from C and H₂, and combustion of C, H₂, CH₄]

(a)

Calculate the enthalpy change of formation of methane, given the enthalpy changes of combustion: C = -394, H₂ = -286, CH₄ = -890 kJ mol⁻¹.

[3]

Question Pack

TOPIC 6 – TOPIC 6: ELECTROCHEMISTRY

121 Electrolysis – 9701/22/O/N/20/Q2(b) Syllabus 6.2

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

122 Redox Processes – 9701/23/F/M/21/Q1(a) Syllabus 6.1

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

123 Electrolysis – 9701/24/M/J/22/Q2(b) Syllabus 6.2

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

124 Redox Processes – 9701/21/O/N/23/Q1(a) Syllabus 6.1

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

125 Electrolysis – 9701/22/F/M/24/Q2(b) Syllabus 6.2

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

126 Redox Processes – 9701/23/M/J/19/Q1(a) Syllabus 6.1

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

Question Pack

127 Electrolysis — 9701/24/O/N/20/Q2(b) Syllabus 6.2

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

128 Redox Processes — 9701/21/F/M/21/Q1(a) Syllabus 6.1

What is the oxidation number of sulfur in H_2SO_4 ?

(a)

Select the correct option.

A +2

B +4

C +6

D -2

[1]

129 Electrolysis — 9701/22/M/J/22/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 18 A flows for 30 minutes. ($A_r = 59$, charge = +2).

[3]

130 Redox Processes — 9701/23/O/N/23/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 20 A flows for 30 minutes. ($A_r = 60$, charge = +2).

[3]

131 Electrolysis — 9701/24/F/M/24/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 22 A flows for 30 minutes. ($A_r = 61$, charge = +2).

[3]

132 Redox Processes — 9701/21/M/J/19/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 24 A flows for 30 minutes. ($A_r = 62$, charge = +2).

[3]

133 Electrolysis — 9701/22/O/N/20/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 26 A flows for 30 minutes. ($A_r = 63$, charge = +2).

[3]

134 Redox Processes — 9701/23/F/M/21/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 28 A flows for 30 minutes. ($A_r = 64$, charge = +2).

[3]

Question Pack

135 Electrolysis – 9701/24/M/J/22/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 30 A flows for 30 minutes. (Ar = 65, charge = +2).

[3]

136 Redox Processes – 9701/21/O/N/23/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 32 A flows for 30 minutes. (Ar = 66, charge = +2).

[3]

137 Electrolysis – 9701/22/F/M/24/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 34 A flows for 30 minutes. (Ar = 67, charge = +2).

[3]

138 Redox Processes – 9701/23/M/J/19/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 36 A flows for 30 minutes. (Ar = 68, charge = +2).

[3]

139 Electrolysis – 9701/24/O/N/20/Q2(b) Syllabus 6.2

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 38 A flows for 30 minutes. (Ar = 69, charge = +2).

[3]

140 Redox Processes – 9701/21/F/M/21/Q1(a) Syllabus 6.1

A molten salt is electrolysed.

(a)

Calculate the mass of metal deposited when a current of 40 A flows for 30 minutes. (Ar = 70, charge = +2).

[3]

Question Pack

TOPIC 7 — TOPIC 7: EQUILIBRIA

141 Chemical Equilibria — 9701/22/O/N/20/Q3(c) Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

142 Chemical Equilibria — 9701/23/F/M/21/Q3(c) Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

143 Chemical Equilibria — 9701/24/M/J/22/Q3(c) Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

144 Chemical Equilibria — 9701/21/O/N/23/Q3(c) Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

145 Chemical Equilibria — 9701/22/F/M/24/Q3(c) Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

146 Chemical Equilibria — 9701/23/M/J/19/Q3(c) Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ? [1]

A High T, Low P

B Low T, High P

C Low T, Low P

D High T, High P

Question Pack

147 Chemical Equilibria — 9701/24/O/N/20/Q3(c) Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ?

- A High T, Low P
- B Low T, High P
- C Low T, Low P
- D High T, High P

[1]**148 Chemical Equilibria — 9701/21/F/M/21/Q3(c)** Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ?

- A High T, Low P
- B Low T, High P
- C Low T, Low P
- D High T, High P

[1]**149 Chemical Equilibria — 9701/22/M/J/22/Q3(c)** Syllabus 7.2

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ?

- A High T, Low P
- B Low T, High P
- C Low T, Low P
- D High T, High P

[1]**150 Chemical Equilibria — 9701/23/O/N/23/Q3(c)** Syllabus 7.1

[DIAGRAM: Haber process equilibrium curve]

(a)Which condition increases the yield of NH_3 ?

- A High T, Low P
- B Low T, High P
- C Low T, Low P
- D High T, High P

[1]**151 Chemical Equilibria — 9701/24/F/M/24/Q3(c)** Syllabus 7.2Consider the dynamic equilibrium: $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ **(a)**

Define the term dynamic equilibrium.

[2]**(b)**

Explain the effect of adding a catalyst on the position of equilibrium.

[2]**152 Chemical Equilibria — 9701/21/M/J/19/Q3(c)** Syllabus 7.1Consider the dynamic equilibrium: $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ **(a)**

Define the term dynamic equilibrium.

[2]**(b)**

Explain the effect of adding a catalyst on the position of equilibrium.

[2]

Question Pack

153 Chemical Equilibria — 9701/22/O/N/20/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

154 Chemical Equilibria — 9701/23/F/M/21/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

155 Chemical Equilibria — 9701/24/M/J/22/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

156 Chemical Equilibria — 9701/21/O/N/23/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

157 Chemical Equilibria — 9701/22/F/M/24/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

158 Chemical Equilibria — 9701/23/M/J/19/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

159 Chemical Equilibria — 9701/24/O/N/20/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

160 Chemical Equilibria — 9701/21/F/M/21/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

Question Pack

161 Chemical Equilibria – 9701/22/M/J/22/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

162 Chemical Equilibria – 9701/23/O/N/23/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

163 Chemical Equilibria – 9701/24/F/M/24/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

164 Chemical Equilibria – 9701/21/M/J/19/Q3(c) Syllabus 7.1

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

165 Chemical Equilibria – 9701/22/O/N/20/Q3(c) Syllabus 7.2

Consider the dynamic equilibrium: $A + B \rightleftharpoons C + D$

(a) Define the term dynamic equilibrium. [2]

(b) Explain the effect of adding a catalyst on the position of equilibrium. [2]

Question Pack

TOPIC 8 — TOPIC 8: REACTION KINETICS

166 Rates of Reaction — 9701/22/O/N/20/Q4(d) Syllabus 8.2

Fig 1.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

167 Rates of Reaction — 9701/23/F/M/21/Q4(d) Syllabus 8.3

Fig 2.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

168 Rates of Reaction — 9701/24/M/J/22/Q4(d) Syllabus 8.1

Fig 3.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

169 Rates of Reaction — 9701/21/O/N/23/Q4(d) Syllabus 8.2

Fig 4.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

170 Rates of Reaction — 9701/22/F/M/24/Q4(d) Syllabus 8.3

Fig 5.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

171 Rates of Reaction — 9701/23/M/J/19/Q4(d) Syllabus 8.1

Fig 6.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

Question Pack

172 Rates of Reaction — 9701/24/O/N/20/Q4(d) Syllabus 8.2

Fig 7.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

173 Rates of Reaction — 9701/21/F/M/21/Q4(d) Syllabus 8.3

Fig 8.1 shows a Boltzmann distribution curve.

- (a) What happens to the peak of the curve at a higher temperature? [1]
- A Moves left and higher
 - B Moves right and lower
 - C Moves right and higher
 - D Moves left and lower

174 Rates of Reaction — 9701/22/M/J/22/Q4(d) Syllabus 8.1

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

175 Rates of Reaction — 9701/23/O/N/23/Q4(d) Syllabus 8.2

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

176 Rates of Reaction — 9701/24/F/M/24/Q4(d) Syllabus 8.3

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

177 Rates of Reaction — 9701/21/M/J/19/Q4(d) Syllabus 8.1

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

178 Rates of Reaction — 9701/22/O/N/20/Q4(d) Syllabus 8.2

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

179 Rates of Reaction — 9701/23/F/M/21/Q4(d) Syllabus 8.3

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

Question Pack

180 Rates of Reaction — 9701/24/M/J/22/Q4(d) Syllabus 8.1

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

181 Rates of Reaction — 9701/21/O/N/23/Q4(d) Syllabus 8.2

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

182 Rates of Reaction — 9701/22/F/M/24/Q4(d) Syllabus 8.3

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

183 Rates of Reaction — 9701/23/M/J/19/Q4(d) Syllabus 8.1

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

184 Rates of Reaction — 9701/24/O/N/20/Q4(d) Syllabus 8.2

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

185 Rates of Reaction — 9701/21/F/M/21/Q4(d) Syllabus 8.3

The rate of a reaction was measured at different temperatures.

- (a) Use the Boltzmann distribution to explain why a small increase in temperature leads to a large increase in reaction rate. [3]

Question Pack

TOPIC 9 – TOPIC 9: THE PERIODIC TABLE: CHEMICAL PERIODICITY

186 Periodicity – 9701/22/O/N/20/Q5(a) Syllabus 9.2

Fig 1.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

187 Periodicity – 9701/23/F/M/21/Q5(a) Syllabus 9.3

Fig 2.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

188 Periodicity – 9701/24/M/J/22/Q5(a) Syllabus 9.1

Fig 3.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

189 Periodicity – 9701/21/O/N/23/Q5(a) Syllabus 9.2

Fig 4.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

190 Periodicity – 9701/22/F/M/24/Q5(a) Syllabus 9.3

Fig 5.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

191 Periodicity – 9701/23/M/J/19/Q5(a) Syllabus 9.1

Fig 6.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

Question Pack

192 Periodicity — 9701/24/O/N/20/Q5(a) Syllabus 9.2

Fig 7.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

193 Periodicity — 9701/21/F/M/21/Q5(a) Syllabus 9.3

Fig 8.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

194 Periodicity — 9701/22/M/J/22/Q5(a) Syllabus 9.1

Fig 9.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

195 Periodicity — 9701/23/O/N/23/Q5(a) Syllabus 9.2

Fig 10.2 shows the melting points of Period 3 elements.

- (a) Which element has a giant covalent structure? [1]
- A Na
 - B Mg
 - C Si
 - D P

196 Periodicity — 9701/24/F/M/24/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

- (a) Write an equation for the reaction of sodium with oxygen. [1]
- (b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

197 Periodicity — 9701/21/M/J/19/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

- (a) Write an equation for the reaction of sodium with oxygen. [1]
- (b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

Question Pack

198 Periodicity – 9701/22/O/N/20/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

199 Periodicity – 9701/23/F/M/21/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

200 Periodicity – 9701/24/M/J/22/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

201 Periodicity – 9701/21/O/N/23/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

202 Periodicity – 9701/22/F/M/24/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

203 Periodicity – 9701/23/M/J/19/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

204 Periodicity – 9701/24/O/N/20/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

205 Periodicity – 9701/21/F/M/21/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

Question Pack

206 Periodicity – 9701/22/M/J/22/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

207 Periodicity – 9701/23/O/N/23/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

208 Periodicity – 9701/24/F/M/24/Q5(a) Syllabus 9.3

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

209 Periodicity – 9701/21/M/J/19/Q5(a) Syllabus 9.1

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

210 Periodicity – 9701/22/O/N/20/Q5(a) Syllabus 9.2

Period 3 elements react with oxygen to form oxides.

(a) Write an equation for the reaction of sodium with oxygen. [1]

(b) Describe the acid-base character of Al_2O_3 and provide an equation to support your answer. [3]

Mark Scheme

MARK SCHEME – WEEK 1

Q	MARKING POINTS	MARKS
1(a)	B (1)	1
2(a)	B (1)	1
3(a)	B (1)	1
4(a)	B (1)	1
5(a)	B (1)	1
6(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
7(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
8(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
9(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
10(a)	Atoms of the same element/same number of protons (1); with different numbers of neutrons/different mass number (1).	2
11(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
11(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
12(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
12(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
13(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
13(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
14(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
14(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
15(a)	Nuclear charge remains the same but electron being removed is from a more positive ion (1); so electrostatic attraction between nucleus and outer electron increases (1).	2
15(b)	Group 14 (or 4) due to large jump after 4th IE (1).	1
16(a)	A (1)	1
17(a)	A (1)	1
18(a)	A (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
19(a)	A (1)	1
20(a)	A (1)	1
21(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
22(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
23(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
24(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
25(a)	Nuclear charge increases (1); shielding remains similar / electrons added to same shell (1); nuclear attraction for outer electrons increases (1).	3
26(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
27(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
28(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
29(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
30(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
31(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
32(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
33(a)	moles = $1.20 / 24.0$ (1); = 0.0500 mol (1).	2
34(a)	Moles of NaOH = $25.0 \times 0.100 / 1000$ = 0.0025 mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2$ = 0.00125 mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0)$ = 0.0625 mol dm ⁻³ (1).	3
35(a)	Moles of NaOH = $25.0 \times 0.100 / 1000$ = 0.0025 mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2$ = 0.00125 mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0)$ = 0.0625 mol dm ⁻³ (1).	3
36(a)	Moles of NaOH = $25.0 \times 0.100 / 1000$ = 0.0025 mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2$ = 0.00125 mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0)$ = 0.0625 mol dm ⁻³ (1).	3
37(a)	Moles of NaOH = $25.0 \times 0.100 / 1000$ = 0.0025 mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2$ = 0.00125 mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0)$ = 0.0625 mol dm ⁻³ (1).	3
38(a)	Moles of NaOH = $25.0 \times 0.100 / 1000$ = 0.0025 mol (1); Moles of H ₂ SO ₄ = $0.0025 / 2$ = 0.00125 mol (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0)$ = 0.0625 mol dm ⁻³ (1).	3

Mark Scheme

MARK SCHEME – WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
39(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025 \text{ mol}$ (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125 \text{ mol}$ (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625 \text{ mol dm}^{-3}$ (1).	3
40(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025 \text{ mol}$ (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125 \text{ mol}$ (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625 \text{ mol dm}^{-3}$ (1).	3
41(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025 \text{ mol}$ (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125 \text{ mol}$ (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625 \text{ mol dm}^{-3}$ (1).	3
42(a)	Moles of NaOH = $25.0 \times 0.100 / 1000 = 0.0025 \text{ mol}$ (1); Moles of H ₂ SO ₄ = $0.0025 / 2 = 0.00125 \text{ mol}$ (1); Conc H ₂ SO ₄ = $0.00125 \times (1000 / 20.0) = 0.0625 \text{ mol dm}^{-3}$ (1).	3
43(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
44(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
45(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
46(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
47(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
48(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
49(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
50(a)	C: $85.7/12.0 = 7.14$, H: $14.3/1.0 = 14.3$ (1); Ratio 1:2, formula CH ₂ (1).	2
51(a)	Trigonal pyramidal (1); 107 degrees (1).	2
52(a)	Trigonal pyramidal (1); 107 degrees (1).	2
53(a)	Trigonal pyramidal (1); 107 degrees (1).	2
54(a)	Trigonal pyramidal (1); 107 degrees (1).	2
55(a)	Trigonal pyramidal (1); 107 degrees (1).	2
56(a)	Trigonal pyramidal (1); 107 degrees (1).	2
57(a)	Trigonal pyramidal (1); 107 degrees (1).	2
58(a)	Trigonal pyramidal (1); 107 degrees (1).	2
59(a)	Two double covalent bonds (1); between C and O atoms (1).	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
60(a)	Two double covalent bonds (1); between C and O atoms (1).	2
61(a)	Two double covalent bonds (1); between C and O atoms (1).	2
62(a)	Two double covalent bonds (1); between C and O atoms (1).	2
63(a)	Two double covalent bonds (1); between C and O atoms (1).	2
64(a)	Two double covalent bonds (1); between C and O atoms (1).	2
65(a)	Two double covalent bonds (1); between C and O atoms (1).	2
66(a)	Two double covalent bonds (1); between C and O atoms (1).	2
67(a)	Two double covalent bonds (1); between C and O atoms (1).	2
68(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
69(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
70(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
71(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
72(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
73(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
74(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
75(a)	Water has hydrogen bonding between molecules (1); H ₂ S has permanent dipole-dipole forces / van der Waals (1); hydrogen bonds are stronger than permanent dipole-dipole forces (1).	3
76(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
77(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
78(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
79(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3

Mark Scheme

MARK SCHEME – WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
80(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
81(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
82(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
83(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
84(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
85(a)	$pV = nRT$ (1); $n = (1.01 \times 10^5 \times 0.050) / (8.31 \times 300)$ (1); $n = 2.03 \text{ mol}$ (1).	3
86(a)	High temperature (1); Low pressure (1).	2
87(a)	High temperature (1); Low pressure (1).	2
88(a)	High temperature (1); Low pressure (1).	2
89(a)	High temperature (1); Low pressure (1).	2
90(a)	High temperature (1); Low pressure (1).	2
91(a)	High temperature (1); Low pressure (1).	2
92(a)	High temperature (1); Low pressure (1).	2
93(a)	High temperature (1); Low pressure (1).	2
94(a)	High temperature (1); Low pressure (1).	2
95(a)	High temperature (1); Low pressure (1).	2
96(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
97(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
98(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
99(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
100(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
101(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
102(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
103(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
104(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
105(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
106(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
107(a)	Enthalpy change when one mole of a compound (1); is formed from its elements (1); under standard conditions in their standard states (1).	3
108(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
109(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
110(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
111(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
112(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
113(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
114(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
115(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
116(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
117(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
118(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
119(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3
120(a)	$\Delta H_f = \sum(\Delta H_c \text{ reactants}) - \sum(\Delta H_c \text{ products})$ (1); = $(-394) + 2(-286) - (-890)$ (1); = -76 kJ mol^{-1} (1).	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
121(a)	C (1)	1
122(a)	C (1)	1
123(a)	C (1)	1
124(a)	C (1)	1
125(a)	C (1)	1
126(a)	C (1)	1
127(a)	C (1)	1
128(a)	C (1)	1
129(a)	$Q = It = 18 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
130(a)	$Q = It = 20 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
131(a)	$Q = It = 22 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
132(a)	$Q = It = 24 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
133(a)	$Q = It = 26 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
134(a)	$Q = It = 28 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
135(a)	$Q = It = 30 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
136(a)	$Q = It = 32 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
137(a)	$Q = It = 34 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
138(a)	$Q = It = 36 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
139(a)	$Q = It = 38 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
140(a)	$Q = It = 40 \times 1800$ (1); moles of $e^- = Q/96500$ (1); mass = moles/2 * Ar (1)	3
141(a)	B (1)	1
142(a)	B (1)	1
143(a)	B (1)	1

Mark Scheme

MARK SCHEME – WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
144(a)	B (1)	1
145(a)	B (1)	1
146(a)	B (1)	1
147(a)	B (1)	1
148(a)	B (1)	1
149(a)	B (1)	1
150(a)	B (1)	1
151(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
151(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
152(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
152(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
153(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
153(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
154(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
154(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
155(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
155(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
156(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
156(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
157(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
157(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
158(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
158(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
159(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
159(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
160(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
160(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
161(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
161(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
162(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
162(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
163(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
163(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
164(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
164(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
165(a)	Rate of forward reaction equals rate of backward reaction (1); concentrations of reactants and products remain constant (1)	2
165(b)	No effect on position of equilibrium (1); increases rate of forward and backward reactions equally (1)	2
166(a)	B (1)	1
167(a)	B (1)	1
168(a)	B (1)	1
169(a)	B (1)	1
170(a)	B (1)	1
171(a)	B (1)	1
172(a)	B (1)	1
173(a)	B (1)	1
174(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
175(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
176(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
177(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
178(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
179(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
180(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
181(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
182(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
183(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
184(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
185(a)	More particles have energy $\geq$ activation energy (1); more successful collisions per unit time (1); higher frequency of collisions (1)	3
186(a)	C (1)	1
187(a)	C (1)	1
188(a)	C (1)	1
189(a)	C (1)	1
190(a)	C (1)	1
191(a)	C (1)	1
192(a)	C (1)	1
193(a)	C (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
194(a)	C (1)	1
195(a)	C (1)	1
196(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
196(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
197(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
197(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
198(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
198(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
199(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
199(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
200(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
200(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
201(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
201(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
202(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
202(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
203(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
203(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
204(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
204(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
205(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
205(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
206(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1

Mark Scheme

MARK SCHEME — WEEK 1 (CONTINUED)

Q	MARKING POINTS	MARKS
206(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
207(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
207(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
208(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
208(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
209(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
209(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3
210(a)	$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ (1)	1
210(b)	Amphoteric (1); reacts with acid: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ (1); reacts with base: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ (1)	3